LEXICAL VARIABLE SELECTION VIA KEYWORDS

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Abstract

This document provides a brief explanation about lexical variable selection via keywords in a typical Lexical Multi-Dimensional Analysis (LMDA) (Berber Sardinha & Fitzsimmons-Doolan, 2025).

Keywords: Lexical Multi-Dimensional Analysis; keywords

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1 Lexical Multi-Dimensional Analysis Overview

Lexical multidimensional analysis (LMDA), an extension of the multidimensional (MD) analysis framework developed by Biber in the 1980s ("multi-feature multidimensional analysis") to study register variation. Through the identification of (lexical) dimensions or sets of correlated lexical features, LMDA enables the analysis of lexical patterning from a multidimensional perspective. These lexical dimensions represent a variety of latent, macro-level discursive constructs. (Berber Sardinha & Fitzsimmons-Doolan, 2025, p.1)

A typical LMDA comprises the procedures described in Figure 1.

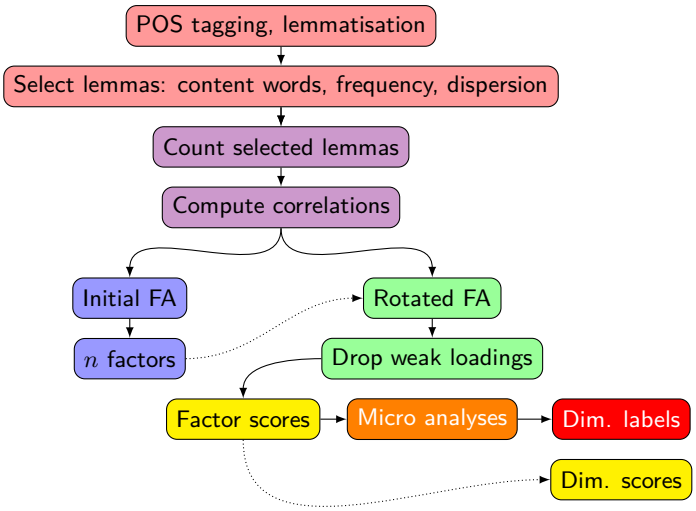


Figure 1: Lexical Multi-Dimensional Analysis Procedures

2 Lexical Variable Selection via Keywords

Figure 2 describes the selection of lexical variables.

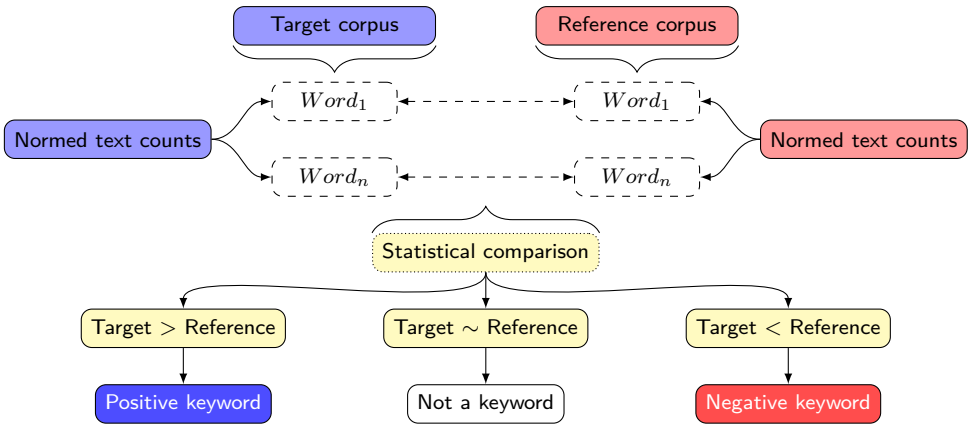


Figure 2: Lexical variables selection via keywords (Scott, 1997)

This process is implemented with two programmes in the LMDA pipeline:

- *keylemmas.py*
- *select_kws_stratified.py*

2.1 *keylemmas.py*

keylemmas.py computes key lemmas from the tagged corpus.

The programme compares each subcorpus against all other subcorpora combined and produces a ranked key-lemma table for each subcorpus.

The output is used by *select_kws_stratified.py* to choose a balanced list of positive keywords across subcorpus strata.

Click *Extract key lemmas by group* for an example.

2.2 *select_kws_stratified.py*

select_kws_stratified.py selects a balanced set of positive keywords from subcorpus-specific key-lemma tables.

It reads the key-lemma tables produced by *keylemmas.py*, extracts lemmas classified as positive keywords ('POSKW'), applies filtering rules, enforces an equal maximum quota per subcorpus, and writes:

1. one selected keyword list per subcorpus
2. one consolidated deduplicated keyword list for downstream LMDA processing

Click *Select a stratified keyword set* for an example.

References

Berber Sardinha, T., & Fitzsimmons-Doolan, S. (2025, July 2). *Lexical Multi-Dimensional Analysis: Identifying Discourses and Ideologies*. Cambridge University Press. https://www.cambridge.org/core/elements/lexical-multidimensional-analysis/B2321B62435360F4F7C4AF3F1CCF4E59

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